

Active Recommendation for Email Outreach Dynamics

Čeněk Žid

zidcenek@fit.cvut.cz

Czech Technical University in Prague
Prague, Czech Republic

Rodrigo Alves

rodrigo.alves@fit.cvut.cz

Czech Technical University in Prague
Prague, Czech Republic

Pavel Kordík

pavel.kordik@fit.cvut.cz

Czech Technical University in Prague
Prague, Czech Republic

Abstract

Email outreach remains a cornerstone of modern marketing, enabling direct, timely communication. However, this strategy faces significant personalization challenges, since new campaigns typically lack historical interaction data and rich side information. In this work, we propose a framework that combines collaborative-filtering (CF) signals derived from a shallow autoencoder (SAE) with a Thompson Sampling-based multi-armed bandit to dynamically select small batches of recipients for each email template. We show SAEs help balance exploration and exploitation by quantifying recipient informativeness and confidence, enabling efficient personalization without retraining during active learning. To facilitate reproducibility and future research, we release a large dataset¹ of almost 15 million recipient-message interactions, offering new insights into email outreach dynamics for CF. Our experiments show that our method outperforms multiple baselines in retrieval metrics while retaining interpretable model components.

CCS Concepts

• Information systems → Personalization; • Computing methodologies → Reinforcement learning.

Keywords

Email Outreach; Reinforcement Learning; Shallow Autoencoder

ACM Reference Format:

Čeněk Žid, Rodrigo Alves, and Pavel Kordík. 2025. Active Recommendation for Email Outreach Dynamics. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM '25)*, November 10–14, 2025, Seoul, Republic of Korea. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3746252.3760832>

1 Introduction

Email outreach remains an important channel in marketing campaigns [7, 32, 34], enabling businesses to connect directly with their customers, disseminate timely information, and drive sales. However, the economic burden of distributing large volumes of emails can be significant, particularly when campaigns target broad audiences without careful segmentation [4, 5]. Beyond economic costs,

sending unsolicited or irrelevant messages can be counterproductive to marketing goals, damaging a company’s reputation as recipients may view such communications as intrusive or spam [18, 22].

A promising solution is to use recommendation frameworks for personalized campaigns [17, 30, 32]. However, this approach faces several challenges when considering practical email outreach dynamics: (1) every new campaign suffers from a *cold-start scenario*, limiting the use of collaborative filtering (CF) techniques; (2) rich side information about recipients (e.g., detailed profiles) is often missing, and emails themselves are highly unstructured (e.g., containing attachments or free-form text); and (3) data protection regulations (e.g., GDPR [26] or CAN-SPAM [3]) severely limit the types of personal information that can be legally collected and used.

Literature Review. To address this challenging scenario, **bridging CF with reinforcement learning** has been extensively studied in the literature. For instance, [41] was among the first to frame cold-start as a multi-armed bandit (MAB) problem [1, 28] on top of probabilistic matrix factorization (MF). [11] introduced a fully online Bayesian MF approach, which maintains a posterior distribution over user and item latent factors via a particle filter. With a similar online philosophy, [37] incrementally updates low-rank user and item embeddings with each feedback and employs a UCB-style policy to minimize regret in interactive recommendation. Moreover, BiUCB [38] treats users and items as dual arms; [8] relies on a heteroskedastic Bayesian MF model to quantify prediction uncertainty; and [42] further augments active learning with content features of new items. Finally, complementary strategies exploit higher-level structure: Cluster-Based Bandits [25] pre-cluster users via CF and then treat cluster membership as the arm space, while Thompson Sampling (TS) for cross-domain advertisement recommendation [9] transfers knowledge across multiple bandit tasks (domains).

Another branch of research focuses on **send time optimization (STO)**, which aims to maximize email engagement by delivering messages when recipients are most likely to open them. In this vein, [21] proposed a dynamic online learning solution that leverages clustering to group users with similar behaviors and a TS-based bandit algorithm to identify optimal send times. Likewise, [4] explored using LSTMs for send time prediction, while [27] applied survival models to predict email opens and time-to-open. In the commercial domain, Emarsys (SAP) offers STO [24], claiming increased open rates without impacting unsubscribe rates, while investigates personalized timing [14].

Finally, **optimization of push notifications**, closely related to email STO, also uses RL to maximize long-term value [16, 20, 21]. Frequency control optimization has been studied primarily by large companies – e.g. Facebook [15], LinkedIn [40], Twitter (X) [20], and Duolingo [39], via extensive A/B tests. Some studies also evaluate offline data to mitigate the cost of exploration: recent work

¹<https://github.com/zidcenek/Active-Learning-for-Email-Interaction-Dynamics>



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CIKM '25, Seoul, Republic of Korea

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ACM ISBN 979-8-4007-2040-6/2025/11

<https://doi.org/10.1145/3746252.3760832>

typically focuses on deep RL (e.g., double deep Q-network (DDQN) [40] or deep Q-network [15]) with high accuracy but limited interpretability. A major drawback of these strategies is their reliance on proprietary datasets, which reduces reproducibility. Although the Enron corpus enables analysis of email dynamics [2, 6, 12], it targets internal communication rather than outreach and is outdated, leaving a gap in public datasets for CF in email dynamics.

Main Contributions. We introduce an iterative MAB formulation (with each arm as a recipient) that combines CF signals and TS to personalize cold-start email campaigns without retraining. Our framework works in an active learning strategy, dynamically selecting batches of recipients for each campaign, maximising engagement while efficiently exploring recipient preferences. We adopt a *shallow autoencoder* (SAE) (an emerging CF trend [10, 19, 29, 31, 36]) to drive adaptive TS, providing interpretability and eliminating the need to retrain during the active learning stage. Our main contributions are: (1) An iterative MAB formulation based on TS with SAE-based CF. To our knowledge, we are the first to combine these strategies. (2) A new public dataset of 15 million recipient–message interactions for analyzing email outreach dynamics in a CF and related tasks. (3) Our method achieves state-of-the-art results with the benefit of interpretability.

2 Method

Preliminaries. For a given provider, we observe m recipients and n campaign templates. A template refers to the base structure of an email message, excluding any personalized elements (e.g., customized greetings). As in a typical recommender system scenario, we can model the recipient–template relationship as a matrix $X \in \mathbb{R}^{n \times m}$, where the rows represent templates (items), the columns represent recipients (users), and an entry X_{ij} represents the interaction of recipient $j \in \{1, 2, \dots, m\}$ with template $i \in \{1, 2, \dots, n\}$. Unlike typical recommender systems, we observe the majority of entries in X . We model $X_{ij} = 1$ if the recipient j opened the message related to template i , and $X_{ij} = 0$ otherwise.

Our aim is to predict, before a deadline T , the set of recipients who will open a message related to template $n + 1$, for which no interactions have been observed yet. The deadline T , set by the provider, defines the timeframe within which a batch of messages should be scheduled and sent, taking into account business rules such as a sale period, promotional event, or other time-sensitive requirements. During the period T , trials of sending emails can be observed. For any $0 \leq t \leq T$, let $x_{n+1}(t)$ represent the vector of observed interactions for template $n + 1$ in the time interval $[0, t]$, following the same assumptions as those applied to the rows of X . Thus, $x_{n+1}(0)$ equals the zero vector. Finally, let n_t be the maximum between 1 and the number of recipients who open messages based on template $(n + 1)$ in the interval between 0 and t . We define a normalized version of $x_{n+1}(t)$ as $\bar{x}_{n+1}(t) = x_{n+1}(t)/n_t$.

Model. To address this task, we propose an iterative MAB framework, where each arm a_j represents a recipient j , and the score $s_j(t)$ associated with selecting arm a_j is computed using a SAE. In this framework, messages from the target template are sent to b disjoint sets of recipients, each of size $k \ll m$. The first batch is sent at $t = 0$, with subsequent batches sent at intervals of $w = T/b$, allowing

time between batches to fine-tune the informative knowledge of the CF component. Since interactions for template $n + 1$ are initially unobserved, we designed the framework to operate in an active learning setting, selecting recipients whose observed behaviors are most likely to provide informative insights into the dynamics of other users' opening decisions. After the active learning phase, an additional final batch is sent at time T . Therefore, we define

$$s_j(t) = \alpha(\phi_j p_j) + (1 - \alpha)f_j(t) \quad (1)$$

where $\phi_j = \sum_i X_{ij}/n$ is the proportion of observed opened templates by recipient- j in the first n templates, p_j and $f_j(t)$ are metrics derived from a SAE and α is a hyperparameter. The general score p_j represents an *influential recipient*, with a high value indicating that observing recipient j opening a message provides valuable information about the likelihood of *all other recipients* opening a message. On the other hand, $f_j(t)$ captures the *score specific to recipient j* , based on the observed interactions with template $n + 1$ at time t . More formally, our SAE optimization is defined as:

$$\begin{aligned} \min_{E,D} \quad & \ell(X, \sigma(XB_{E,D})), \\ \text{with } B_{E,D} = & ED^\top - \text{diag}([E \odot D]1_m) \end{aligned} \quad (2)$$

where E and $D \in \mathbb{R}^{n \times d}$ are the encoder and decoder matrices, respectively, d is a tunable hyperparameter that defines the size of the bottleneck layer, 1_m is a unit column vector of size m , $\odot$ represents element-wise multiplication, and $\text{diag}(a)$ represents a diagonal matrix with the diagonal corresponding to the elements of vector a . The loss function $\ell(A, B)$ represents the element-wise binary cross-entropy loss, where $A_{i,j}$ is the ground truth label and $B_{i,j}$ is the predicted probability. Note that in (2), the term $\text{diag}([E \odot D]1_m)$ ensures that the diagonal of matrix $B_{E,D}$ avoids trivial solutions by making the diagonal of $(B_{E,D})$ equal to the zero vector, which is a widely used constraint in SAEs [29, 31, 36].

Remark: The diagonal constraint we impose to be zero, along with modeling $X_{ij} = 0$ when an interaction is not observed – similar to other SAE methods – has a slightly different effect in our case. Observe that in EASE [31], for instance, if a user's interaction is not observed, the resulting predictor would assign a score of zero to all items, leading to complete uncertainty about which item the user is most likely to interact with. Here, however, the sigmoid function ensures that missing interactions do not skew predictions toward extreme values (0 or 1). By assigning a probability of 0.5 when $X_{i,j} = 0, \forall i$, our model follows a neutral prior, reflecting maximum uncertainty rather than assuming a preference.

To better understand the rationale behind our model, we refer to Figure 1. We begin by training our SAE as described in (2) (illustrated on the left-hand side of the figure). Similar to the approach used in shallow autoencoders, an entry (i, j) of the matrix $\Sigma = \sigma(B_{E,D})$ represents the probability that recipient i will open a message from a template, given that only recipient j has opened the message from the same template (see the zoomed-in view in the middle of Figure 1). For instance, in the illustration, the probability of user 1 opening a message from a template that only user 3 has opened is 0.9. Therefore, SAEs can offer several properties that aid in defining scores for the arms in the MAB framework, even in the absence of interactions for a new campaign template.

Our $s_j(t)$ score in (1) is determined by a trade-off between two factors. In the first factor, p_j is computed by averaging the j -th row

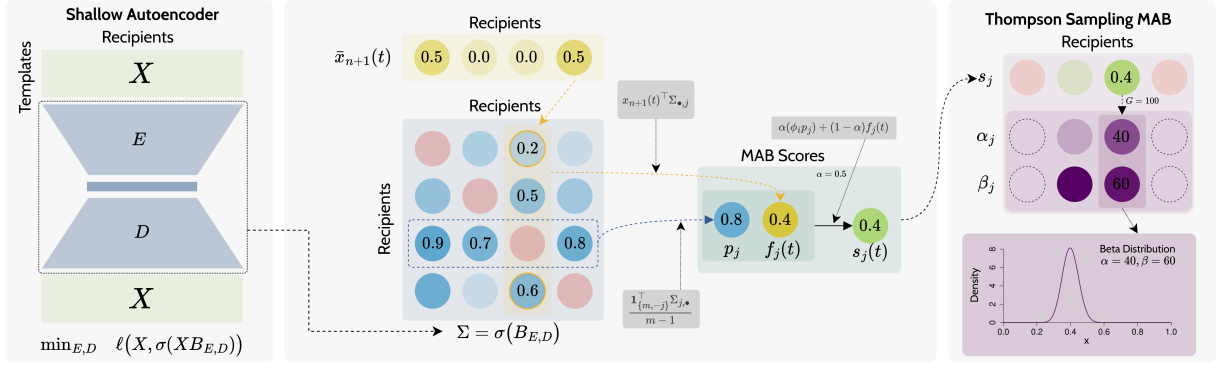


Figure 1: Overview of our proposed method. See Section 2 for details.

of Σ while excluding the j -th element of the row. More formally,

$$p_j = \mathbf{1}_{\{m,-j\}}^\top \Sigma_{j,\bullet} / (m-1), \quad (3)$$

where $\Sigma_{j,\bullet}$ is a column vector containing the elements of the j -th row of Σ , and $\mathbf{1}_{\{m,-j\}}$ is a column vector where all m elements are 1 except the j -th element, which is 0. This score provides a general measure of the impact of recipient j on all other recipients, based on interactions with previous messages, independent of template $n+1$ (see the example of p_j in Figure 1). Thus, even *without observing* any interactions for the new template, the SAE’s matrix Σ provides crucial information for active learning. However, while knowing whether a recipient opens an email could be informative, it may not be the only relevant information when recipient behavior is absent. In some cases, a recipient can be highly informative but not very active. Thus, sending an email to such a recipient would likely result in no interaction, which would have little impact on reducing the model’s uncertainty. To address this, we multiply ϕ_j by p_j , creating a factor that jointly considers both influence and activity. Note that $\lim_{n \rightarrow \infty} \phi_j$ equals the probability of a recipient opening an email from the provider regardless of the content.

The second factor, $f_j(t)$, is computed as:

$$f_j(t) = x_{n+1}(t)^\top \Sigma_{\bullet,j}, \quad (4)$$

and represents the model’s confidence at time t , given the observed interactions specific to the target template $n+1$ in a CF-based setup.

We use a MAB backbone inspired by the TS algorithm [23], which is a probabilistic method that balances exploration/exploitation to maximize rewards over time. For each arm, a Beta distribution models the probability of success, and a sample is drawn from each distribution to select the arm with the highest sampled value. In our approach, we use $s_j(t)$ as the basis for defining the parameters of the Beta distribution in TS, where

$$\alpha_j(t) = G s_j(t) \text{ and } \beta_j(t) = G(1 - s_j(t)). \quad (5)$$

Here, $s_j(t)$ is a score between 0 and 1 that relates to the likelihood of recipient j opening a message, though it is not strictly a probability. The parameter G serves as a *confidence modifier*, scaling the distribution based on the model’s certainty. A higher G indicates stronger confidence in $s_j(t)$, resulting in a narrower Beta distribution with lower variance, while a smaller G reflects greater uncertainty, producing a broader distribution (see the right-hand

side of Figure 1). This formulation allows the model to effectively balance exploration and exploitation: recipients with higher $s_j(t)$ (indicating higher informativeness) are more likely to be selected, while those with lower $s_j(t)$ are explored less frequently.

3 Experimental Setup

Dataset and Baselines. Our experiments were performed on a new Mailprofiler’s **XCampaign dataset** (*XCampaign is an email campaign management platform developed by Mailprofiler*) containing 14 908 085 messages sent over the course of 12 months (with $T = 48$), 131 918 recipients and 160 templates. For each message sent, we collected the recipient ID, the template ID, and (in the case of an open) the time between sending and opening. The open rate is 9.1%. For evaluation, we split the dataset into three chronologically disjoint parts: the last 10 templates for the test set, the preceding 5 for validation, and the remaining templates for training. We do not have access to any kind of side information.

We compare our method with eight baselines, including active learning-based factorization machines **FMFC-DB**, **FMPSAL**, and **FMRSAL** [42]; Q-networks (**DQN** [20]) and deep reinforcement methods for email delivery (**DDQN** [40] and **FactorUCB** [37]). As our model is based on TS [23], we also perform an ablation using the vanilla algorithm, setting α and β according to observed counts of openings. Finally, for a sanity check, **Random**: an algorithm that selects recipients uniformly at random. Each baseline has its own parameter tuning, which we based on the original papers. Given space limitations, for details on reproducibility aspects (such as hyperparameter tuning and optimization of our model and the baselines), please refer to our repository (see the title-page).

Experimental Details: We perform an offline evaluation. Because the interval between sending and opening depends heavily on the time of sending, which cannot be captured in an offline evaluation, we simulate delays: for each opened message, the opening interval is sampled from an exponential distribution with scale equal to the inverse of the original opening interval. We consider scenarios in which the provider limits sending to only a fraction τ of recipients. (e.g., sending 25% implies $\tau = 25$). During active learning, batch size is $k = 0.15 \times m/b$ (15% of messages overall); at time T we send the remaining messages up to the τ bound. If $\tau \leq 15$, all messages are sent during active learning.

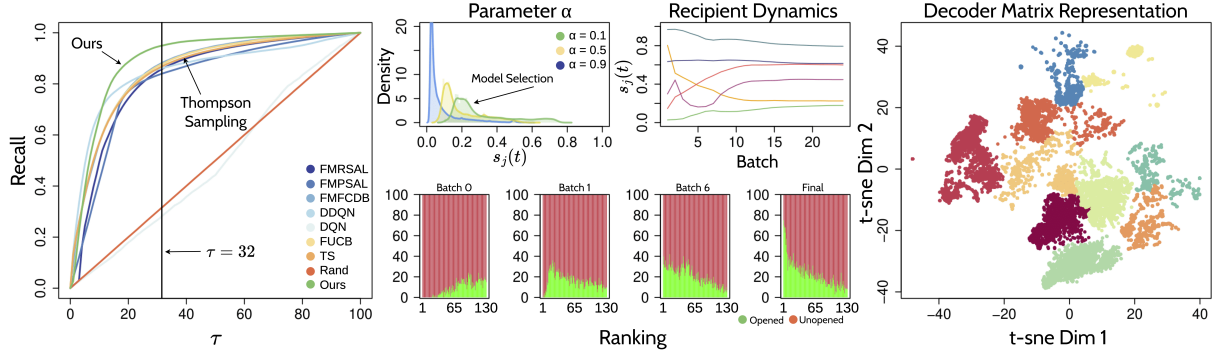


Figure 2: Results and interpretability aspects of our model. See Section 4 for details.

Table 1: Recall performance of our method versus baselines.

Method	Recall@25%	Recall@50%	Recall@75%
FMFC-DB	<u>0.834±0.0002</u>	<u>0.948±0.0001</u>	<u>0.984±0.0001</u>
FMPSAL	0.810±0.0026	0.901±0.0041	0.961±0.0011
FMRSAL	0.805±0.0007	0.931±0.0005	0.975±0.0002
Factor UCB	0.830±0.0001	0.945±0.0001	<u>0.983±0.0001</u>
MBRL	0.226±0.0001	0.443±0.0001	0.792±0.0001
DDQN	<u>0.835±0.0074</u>	0.905±0.0063	0.942±0.0033
TS	0.822±0.0006	0.935±0.0007	0.977±0.0004
Random	0.250±0.0026	0.500±0.0027	0.751±0.0022
Ours	0.923±0.0005	0.975±0.0002	0.989±0.0004

4 Results

Overall Performance: We ran ten simulations as described in Section 3 and report the mean and standard deviation. The results are presented in Table 1 and in the left-hand side of Figure 2. For instance, when $\tau = 25$, the second column of the table (the green line at $\tau = 25$ in the figure) shows opening coverage of 92.3%. Our method outperforms baselines across most target quantiles, especially the most relevant ones. Although DDQN yields better results when sending to fewer than 11% of overall emails, it covers at most 70.89% of openings. Notably, by sending to only 32% of recipients, our method captures 95% of target recipients, substantially reducing the number of emails sent while maintaining high recall.

Active Learning: During the active learning phase, our model sends $b + 1$ batches. The first purely cold-start batch (no prior opening information), while each subsequent batch leverages all openings observed before that sending. The middle panel of Figure 2 shows these results. Based on validation set performance, we selected $b = 24$. The stacked bar plots in the bottom-center panel illustrate how openings and non-openings are distributed across ranking centiles for each batch. The x-axis indicates the ranking centiles (1: positions 1–100, 2: 101–200, ... up to 10% of recipients). For example, in *Batch 0* (pure cold-start), none of the recipients who later opened their emails were placed in the first centile (positions 1–100); by the final batch ($b = 24$), approximately 92% of such recipients appear in the top centile. Intermediate batches further

confirm that active learning accelerates the model’s understanding of opening behavior and that collaborative filtering is effective.

To analyze individual recipients, the *Recipient Dynamics* graph presents the score $s_j(t)$, the primary parameter in our TS framework. Each line represents a sampled recipient’s score $s_j(t)$ across batches. The plot reveals diverse dynamics: for example, the yellow line indicates a recipient whose opening confidence declines over time, whereas the red line shows the opposite trend. We also observe alternating confidence trajectories (purple line) and relatively stable scores in other cases. Furthermore, we analyze the parameter α , which balances the influential score p_j and the temporal recipient-specific score $f_j(t)$, thereby also helping to modulate exploration/exploitation during the active learning phase. In the final batch, we observe that as α decreases (increasing the component $f_j(t)$), the model’s predictions become more confident, yielding higher values of $s_j(t)$ for a specific group of recipients. We selected $\alpha = 0.1$ based on validation.

Finally, a 2D-t-SNE projection [35] of the SAE’s decoder weights D , shown on the right of Figure 2) displays well-defined clusters (each point is a recipient) and CF patterns in recipient–template interactions, consistent with prior work [13, 33, 36]. When combined with our performance results, this projection demonstrates that the SAE-based model effectively integrates CF and MAB strategies, providing substantial benefits in the analyzed domain.

5 Conclusion

We propose an active learning framework for email outreach that combines CF via a shallow autoencoder with a TS-based bandit to select recipient batches without retraining. Despite its simplicity, our method achieves remarkable efficiency—retaining nearly 90% of positive responses while sending only 20% of emails. This reduction enables brands to minimize digital fatigue, reduce environmental costs, and maintain engaged users without inbox saturation. We release a public dataset of nearly 15 million user–email interactions to promote reproducibility and further study which may integrate time-sensitive send-time policies or richer temporal dynamics.

Acknowledgments

Čeněk Žid’s research was supported by the Grant Agency of the Czech Technical University (SGS20/213/OHK3/3T/18). We warmly thank *Mailprofiler* for providing the dataset for this research.

Generative AI Usage Disclosure

Generative AI tools were used in this research to assist in processing data, generating graphical representations such as graphs and charts, and improving the linguistic aspects of this work, including grammatical accuracy, sentence structure, and overall readability. We have reviewed all AI-generated content for accuracy and correctness; the conclusions in this work remain solely the responsibility of the authors.

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